Local gods, supernatural punishment, and the expansion of human sociality: A cross-cultural causal analysis (Preregistration)

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Abstract

Religion has long been considered a significant factor in human cooperation. In particular, beliefs in morally concerned, punitive, and knowledgeable supernatural agents may motivate cooperative behaviour, and potentially facilitate the kinds of widespread cooperation that are uniquely expressed among humans. While previous cross-cultural research is consistent with this idea, many open questions remain, chief among them being the extent to which the so-called "moralistic traditions" – those that espouse beliefs in gods that specifically care about and judge interpersonal social behaviors (e.g., the gods of "world" religious traditions such as Christianity or Hinduism) – enhance cooperation beyond the scope of smaller, more locally confined or indigenous traditions. Here, we assess this question among participants from 22 societies ($n \approx 3{,}000$) across multiple experimental measures of cooperation towards anonymous, geographically distant co-religionists of either "local" or "moralistic" religions, and various ways to operationalise such traditions. Our target estimand is the joint causal effect of supernatural punishment and omniscience beliefs on cooperation with distant co-religionists, and whether this is moderated by tradition type.

Keywords: Supernatural Punishment, Morality, Cooperation, Causality, Cross-Cultural

1 Background

Over the course of human evolution, the scope and scale of human cooperation and sociality has greatly expanded, culminating in modern-day nation states and global cooperative institutions. Many factors have contributed to this increase in social complexity, including agriculture, geography, warfare, technology, and secular institutions, among others (Turchin et al., 2022; Chira et al., 2024; Johnson and Earle, 2000). A long-standing and much-debated proposal is that there is something unique to the so-called "moralistic" or "world" religious traditions partly by virtue of their gods explicitly caring about how we treat other people and meting out punishments in response to said behaviour (Purzycki, 2025b). Moreover, across many fields, recent efforts probe the relationship between these traditions and facets of social complexity (Norenzayan et al., 2016; Baumard and Boyer, 2013; Bellah, 2011; Turchin et al., 2023; Watts et al., 2015). However, while previous studies

have examined the association between individual beliefs and cooperation beyond local communities, the bulk of such studies were conducted among participants associated with moralistic traditions. We therefore ask: Are beliefs regarding deities from "moralistic" religious traditions any more effective at promoting cooperation with unknown and distant co-religionists than beliefs regarding deities from "local" religious traditions?

At face value, these expansive patterns of human cooperation – defined broadly as "a behaviour which evolved to benefit others" (West et al., 2007) – are puzzling from an evolutionary perspective. This is because large-scale cooperation cannot easily be understood by canonical theories of cooperative evolution, such as kin selection or direct reciprocity (Hamilton, 1964; Trivers, 1971). While other factors beyond relatedness and reciprocity may promote cooperation, such as reputation, assortment, punishment, and institutions (Ostrom, 1990; Raihani and Bshary, 2019; Roberts et al., 2021; Cronk and Leech, 2012), religious beliefs may also impact our cooperative tendencies.

A growing body of work highlights the apparent motivating effect of individual belief in punitive and omniscient supernatural agents on cooperation (Norenzayan et al., 2016; Norenzayan, 2013; Purzycki and Sosis, 2022; Purzycki, 2025b; Lang et al., 2024; Purzycki et al., 2024). To understand these patterns, the "supernatural punishment hypothesis" suggests that beliefs in punitive and omniscient deities may increase the perceived benefits of cooperation, the perceived costs of punishment, and perceptions regarding the probability of punishment, thereby helping to promote cooperation beyond traditional kin- or reciprocity-based systems (Johnson and Krüger, 2004; Schloss and Murray, 2011). The extent to which this may be universal across religious traditions, however, is unclear. Some posit a co-evolutionary process between supernatural punishment (and religion more broadly) and the ever-broadening scale of cooperation, giving rise to these "moralistic" religious traditions (Norenzayan et al., 2016). That is, there may be something unique about these particular traditions in motivating cooperative behaviour.

Cross-cultural experiments conducted among 15 non-Western societies have found that belief in punitive and omniscient deities was associated with higher levels of cooperation towards distant co-religionists (Purzycki et al., 2016; Lang et al., 2019). This suggests that belief in monitoring and punitive deities may motivate cooperative behaviour beyond beyond our everyday interaction spheres, to encompass unrelated and unknown co-religionist group members, thus expanding the scale of human cooperation. For a few key reasons, however, these studies are largely silent on the question of whether this varies by tradition (Purzycki et al., 2022a). The present project speaks to three in particular.

First, in the majority of these societies, cooperation was measured as a transaction between individuals of the same moralistic religion (e.g., if supernatural punishment beliefs among Christians promoted cooperation towards other geographically-distant Christians). As the contrast of interest was the presence or absence of beliefs, it remains unclear if cooperation across tradition types is any different. Put another way, it is not known whether beliefs regarding supernatural punishment of local or indigenous – but less explicitly moralistic – deities (e.g., gods that care more about rituals or resource management) may have a similar positive effect on cooperation towards distant co-religionists of the same *local* religious traditions. Second, subsequent studies using the same dataset found that the degree to which individuals ascribe moral concern to deities had little-to-no relationship with cooperation towards distant co-religionists (Bendixen et al., 2023), suggesting that although punitive and omniscience beliefs may motivate behaviour, perceived moral concern may not. Finally, these previous studies did not employ formal causal inference methods (for an exception, see Bendixen et al., 2023), and hence the extent to which results can be given a causal interpretation remains unclear (Hernán and Robins, 2020; Rohrer, 2018; Bulbulia, 2024; Pearl et al.,

2016; McElreath, 2020).

In this report we adopt a causal inference approach to assess whether the relationship between belief in punitive and omniscient deities and cooperation with distant co-religionists differs between local and moralistic religious traditions. To make this comparison, we combine previous studies' data from 15 societies (Lang et al., 2024; Purzycki et al., 2024) with newly-collected data from seven small-scale societies which primarily focused on cooperation within *local* religious traditions. This allows us to test whether any effect of supernatural punishment and monitoring on cooperation with distant co-religionists is specific to moralistic religious traditions, or a feature of religious traditions more generally. Either result is theoretically interesting. If the relationship between supernatural punishment and cooperation is stronger in moralistic religious traditions, this would suggest that there may indeed be something particular about how these religions mobilise cooperation. In contrast, if there is no difference between religion types, this would be evidence against these theories as it would suggest that supernatural punishment may promote cooperation equally, irrespective of religious tradition. More generally, these data also allow us to assess whether *overall* levels of cooperation differ between religious traditions, speaking to broader debates regarding the potential motivating effect of tradition types on cooperation – e.g., that moralistic traditions may promote cooperation more effectively than local ones – in addition to whether these tradition types moderate any effect of supernatural punishment on cooperative behaviour.

2 Methods

This document is a pre-registration of our intended analysis plans. The majority of data has already been collected, and from previous publications (Purzycki et al., 2024; Lang et al., 2024) and auditing of the newly-collected data, we do possess some knowledge about the data we plan to use. This has been useful to understand some of nuances and complexities inherent in such large-scale cross-cultural projects (e.g., slight variations in protocols between sites, levels of missing data, etc.) which help inform our analysis decisions. However, despite having access to this data, we assert that we have not conducted any of the analyses described below and therefore do not know what the results of our analyses will be. This also explains why there are many tense shifts throughout this document; the data has already been collected (hence this is in the past tense), while we are describing our intended analyses which we have not conducted yet (hence this is in the future tense).

2.1 Participants

Our analyses include data from 22 diverse societies ($n \approx 3{,}000$) from a range of subsistence types, including hunter-gatherers, horticulturalists, agriculturalists, pastoralists and market-integrated societies (see Table 1 and Supplementary Table S1 for summaries, and Figure 1 for a map). Data were collected in three phases (for more information on the different phases of data collection, see Supplementary Section 1.1). Participants were recruited based on their actual or default religious affiliation; in the previous rounds of data collection this was predominantly based on belonging to a moralistic world religious tradition (e.g., being Hindu in Lovu, Fiji, or being Buddhist in Tyva), while in the most recent round of data collection this was predominantly based on belonging to a local or indigenous religious tradition (e.g., being a forest-spirit initiate among the BaYaka, or identifying with a Vodun tradition in Benin). In many sites religious affiliation was assumed by default, rather than explicitly asked, as based on ethnographic knowledge all individuals were

known to be of the same religious tradition. All sites collected data from participants within one religious tradition, other than Benin, where participants from both local (Vodun) and moralistic (Christian) religious traditions were included. For a brief ethnographic description of each of these field sites and site-specific information regarding data collection, see Supplementary Section 1.2.

Figure 1: Map of 22 field sites.

Group	Country	*N*	Moralistic God	Local God	Participant	Local	Distant
Altai Uriankhai	Mongolia	86	Buddha	Spirit-masters	Person from Duut *sum*	*Ovoo* practitioner from Duut *sum*	*Ovoo* practitioner from Bayandun *sum*
BaYaka	Republic of Congo	98	*Komba* (Creator deity)	Forest spirits	BaYaka spirit-initiate from camp	BaYaka spirit-initiate from same camp	BaYaka spirit-initiate from distant camp
Buryat	Mongolia	82	Buddha	Spirit-masters	Person from Bayandun *sum*	*Ovoo* practitioner from Bayandun *sum*	*Ovoo* practitioner from Duut *sum*
Cachoeira	Brazil	274	Christian God	Candomblé God (*Ogum)*	Candomblé member in Cachoeira	Candomblé member in Cachoeira	Candomblé member in Salvador
Coastal Benin (Christian)	Benin	91	Christian God	Vodun spirits	Self-identified Christian believer	Christian believer from same village	Christian believer from different village
Coastal Benin (Vodun)	Benin	256	Christian God	Vodun spirits	Self-identified Vodun believer	Vodun believer from same village	Vodun believer from different village
Coastal Tanna	Vanuatu	178	Christian God	Garden spirit (*Tupunus*)	Person from Christian village	Person from same village	Christian from other village
Hadza	Tanzania	201	*Haine* (Traditional)	*Ishoko* (Traditional)	Person from Hadza camp	Hadza from same camp	Hadza from different camp
Huatasani	Peru	94	Christian God	Mountain spirits *apus* or Catholic saints	Person from Huatasani or nearby village	Catholic from same town/village	Catholic from Quechua Taraco/ Aymara Vilque Chico
Indo-Mauritians	Mauritius	245	Hindu *Shiva*	Ghost (*Nam*)	Hindu from La Gaulette	Hindu from La Gaulette	Hindu from Point aux Piments
Inland Tanna	Vanuatu	112	*Kalpanan* (Traditional)	Garden spirit (*Tupunus*)	Person from Kastom village	Person from same village	Person from other Kastom village
Kananga	Democratic Republic of Congo	200	Christian God	Ancestor spirits	Born-again Luluwa from Kananga	Born-again Luluwa from Kananga	Born-again non-Luluwa from Kananga
Kichwa	Ecuador	85	Catholic God	Pachamama	Person from Yawari village/ Rukullakta pueblo	Pachamama believer from Yawari/Rukullakta	Pachamama believer from Palimbe
Lovu	Fiji	76	Hindu *Bhagwan*	-	Hindu from Lovu	Hindu from Lovu	Hindu from other Fijian island
Marajó	Brazil	77	Christian God	Virgin Mary	Evangelical/Catholic from Marajó	Evangelical/Catholic from Marajó	Evangelical/Catholic from Rondon
Mysore	India	165	Hindu *Shiva*	*Chamundeshwari* (Local goddess)	Hindu from Mysore	Person from same neighborhood	Hindu from Tamil Nadu
Samburu	Kenya	40	Christian God/ Traditional (*Nkai*)	-	Person from Mbaringon	Person from Mbaringon	Samburu with similar beliefs from Poro
Sihanaka	Madagascar	79	*Ndrianampanjaka* (Divine ruler)	*Zazavavindrano* (Water spirits)	Person from Anororo	Person from Anororo who cultivates relations with *zazavavindrano*	Person from Ambohidava who cultivates relations with *zazavavindrano*
Sursurunga	Papua New Guinea	163	Christian God	*Sirmát* (Spirit)	Person from Nokon	Person from clan	Person from church
Toraja	Indonesia	80	*Puang Matua* (Traditional/Christian)	Ancestor spirits	Person from Puangbembe Mesakada	*Aluk to Dolo* adherent from Puangbembe Mesakada	*Aluk to Dolo* adherent from village in Mappak district
Turkana	Kenya	247	Christian God	Ancestor spirits	Person from Nadome	Turkana who attends nearby church	Turkana who attends church in Awar Naparan
Tyva	Russia	81	Buddha	Spirit-masters	Buddhist from Kyzyl	Buddhist from Kyzyl	Buddhist from Ak Dovurak
Yasawa	Fiji	75	Christian God	Ancestor spirits	Person from Yasawa	Person from Yasawa	Church-attender from different island

Table 1: Summary description of 22 study sites. Local and Distant columns describe recipients in experiments.

2.2 Data

For the majority of sites, data collection occurred in two steps, with the first informing the design of the second. First, there was an initial "Religious Landscape Interview," which consisted of qualitative and quantitative interviews with approximately 20 participants to understand the religious context and select the deities to be the moralistic god – a punitive and monitoring deity (most often a god from a world religion such as Christianity, Buddhism or Hinduism) – and a local god – ideally a less punitive and less knowledgeable deity which was less morally-concerned with interpersonal behaviour (usually, but not always, an "indigenous" deity outside a world religion). After this initial interview, the second and main phase of data collection consisted of: i) demographic data; ii) religiosity survey data on the moralistic and local gods; and iii) experimental games. All participants (ideally a minimum of 80 participants per site) played the experimental games before the interviews. Any site-specific deviations from this core protocol are discussed in Supplementary Section 1.3.

2.2.1 Experimental games

To assess cooperation between geographically distant co-religionists who were unlikely to have reciprocal transactions, we used two experimental economic games; the Random Allocation Game (RAG; Hruschka et al., 2014) and the Dictator Game (DG; Camerer, 2003). We conducted two versions of each game; one between the participant and a randomly selected, anonymous, and geographically distant co-religionist (SELF vs. DISTANT condition), and the other between a local co-religionist and another distant co-religionist individual (LOCAL vs. DISTANT condition). For each version of each game, participants were shown two opaque cups (or equivalent), representing the dyads above (i.e., SELF vs. DISTANT, or LOCAL vs. DISTANT), to which they had to allocate money (or tokens representing money) as dictated by the rules of the games. In some sites, the co-religionists were part of the moralistic religious tradition, while in others they were from a local tradition (Table 1). All sites played RAGs, with 18 also having played DGs. When both games were played in the same session, RAGs were conducted before the DGs, with the order of conditions counter-balanced across games. Otherwise, participants played RAGs and DGs approximately one year apart.

In the RAGs, participants were given a fair two-coloured die. Prior to rolling the die, they were asked to think about which cup to put a coin into. After rolling, if the die came up a pre-specified colour (e.g., black), they were told to put the coin in the cup they were thinking of; but if it came up the other colour (e.g., white), they were told to put the coin in the other cup. Participants played alone, so could violate these rules and bias their allocations to whichever cup they preferred. There were a total of 30 die rolls for each RAG. In the DG, participants were simply given ten coins to divide between the cups as they wished.

Prior to playing the RAGs, participants answered comprehension check questions to ensure that they understood the rules of the games; participants who did not pass these comprehension checks did not play the games. Each allocation of money was the equivalent of approximately 2% of an average daily wage in the country or local economy (i.e., for both RAGs together, the total stakes were roughly an average daily wage). Money was distributed to participants, randomly-chosen local co-religionists, and randomly-chosen distant co-religionists after the games had been played. Participants also received a show-up fee of approximately 25% of an average daily wage.

2.2.2 Religiosity interview

The religiosity interview consisted of a battery of questions related to individuals' beliefs about, and practices towards, the designated moralistic and local gods. We randomized the order of the god-specific questions across participants. The core religiosity questions used here concerned "punishment" and "knowledge breath" (i.e., omniscience, or monitoring).

We asked two questions regarding punishment: "Does [deity] ever punish people for their behaviour?" and "Can [deity] influence what happens to people after they die?" Both questions were binary yes/no variables. Consistent with previous work (Purzycki et al., 2016; Lang et al., 2019), we will average these two variables together. This is for both practical (to limit the number of interaction terms; see below) and theoretical reasons (as these questions have been previously found to measure the same/similar construct).

We also asked two questions regarding supernatural monitoring: "Can [deity] see into people's hearts or know their thoughts and feelings?" and "Can [deity] see what people are doing if they are far away in XXXX [a distant town/city]?" As above, both of these questions were binary variables which will be averaged together for analyses.

For each society, we will use supernatural belief data from the deity of the same religious tradition as the target co-religionists as our exposure variables. For instance, among Tyvans, co-religionists were Buddhists, so supernatural beliefs regarding Buddha will be our exposures for this site; while among the BaYaka, co-religionists were forest spirit-initiates, so forest spirit punitive and omniscience beliefs will be our exposures for that site.

Other religiosity variables used in analyses include ritual practice, based on the question "Do you perform activities or practices to talk to, or appease, [deity]?" (also a binary question). "Moral concern scales," based on how concerned each deity was with punishing lying, theft and murder, were also asked to participants (Purzycki et al., 2022b). Free-list data on what makes both the moralistic and local gods angry/what they dislike was also collected as part of this religiosity interview (up to five items); this was coded into general themes (see Purzycki and McNamara, 2016), the most relevant of which here is "moral concern," referring to norms or social behaviours with a direct cost or benefit to others (e.g., helping, honesty, sharing, selfishness; Bendixen et al., 2024, 2023).

2.2.3 Demographics

We also asked participants a range of questions related to sociodemographic characteristics. Of most relevance to this study are age, sex, years of formal education, number of children and food security, as they are plausible confounders of our focal relationship (see below). To assess food security, four binary questions were asked regarding worry over having enough food in the next month, 6 months, year, and 5 years; these were averaged together to give a score from "0" (no food insecurity) to "1" (high food insecurity).

2.2.4 Operationalising Moralistic and Local Religious Traditions

To assess the robustness of our constructs, we operationalise "moralistic" and "local" traditions in a variety of ways. First, we will group so-called "world religions" – such as Christianity, Buddhism and Hinduism – as "moralistic" traditions and all other religious traditions as "local." Second, we will use a "relative" (i.e., within-site) definition of moralistic traditions operationalised as "the more punitive, omniscient and morally concerned of the two selected deities." We will determine this in two ways: i) based on researchers' selected deities; and ii) a data-driven approach comparing

these two deities within each site using the punishment mini-scales, omniscience mini-scales, moral concern scales, and free-list data on gods' moral concern. Third, we will apply an "absolute" (i.e., between-site) definition using a data-driven approach to categorise societies based on free-list responses regarding the degree to which individuals ascribe moral concern to deities of the religious tradition of the game recipients (Purzycki, 2025a; Major-Smith and Purzycki, 2026); that is, sites with focal deities higher in moral concern compared to other societies will be coded as "moralistic," and sites with deities lower in moral concern as "local." Despite these different operationalisations, in most cases determining the moralistic and local tradition is clear (e.g., Christianity is "moralistic"). As such, any ambiguity only applies to a handful of sites (see Table 1 and Supplementary Table S2). For more discussion on these decisions, see Supplementary Section 1.4.

2.3 Causal Model

Our estimand here is "the average causal effect of supernatural beliefs regarding punishment and omniscience on cooperation with distant co-religionists, and whether this is moderated by local vs. moralistic religious traditions." A Directed Acyclic Graph (DAG) displaying our causal assumptions regarding the data-generating mechanisms is in Figure 2. There are a number of key assumptions embedded in this DAG (further discussion of these points can be found in Supplementary Section 1.5.1, with discussion of additional causal assumptions in Supplementary Section 1.5.2):

1. Our exposure "supernatural beliefs" is the *joint* effect of both beliefs regarding punishment and omniscience of the focal deity (i.e., the moralistic or local god, depending on the religious tradition of the co-religionist recipients).

2. We are interested in whether the effect of the supernatural beliefs exposure is moderated by tradition type (as denoted by the "effect modifier" node).

3. There are a key set of "core confounders" – age, sex, education and food security – which may plausibly cause both the exposure and the outcome (Baimel et al., 2022; Purzycki and Bendixen, 2024; Vardy et al., 2022; Lang et al., 2026).

4. Given that the data are cross-sectional, there is ambiguity as to whether some variables are confounders, mediators, or both (or neither; grouped under the "Confounders and/or mediators" node). Given this uncertainty, we will conduct sensitivity analyses to assess how robust our results are to different causal assumptions – i.e., treating these variables as confounders vs. as mediators (Major-Smith, 2023).

5. There is the risk of unmeasured confounders biasing our estimates (denoted by the node "U").

6. Features of the experimental game set-up may impact cooperation (e.g., order of games, level of comprehension for RAGs), although as these cannot cause our exposure, they are not confounders.

2.4 Statistical Models

Our statistical approach follows directly from our causal assumptions (Figure 2). Our base model has the supernatural punishment and omniscience mini-scales as joint exposures, and the core confounders and game features outlined above as covariates. We will use multi-level models, with

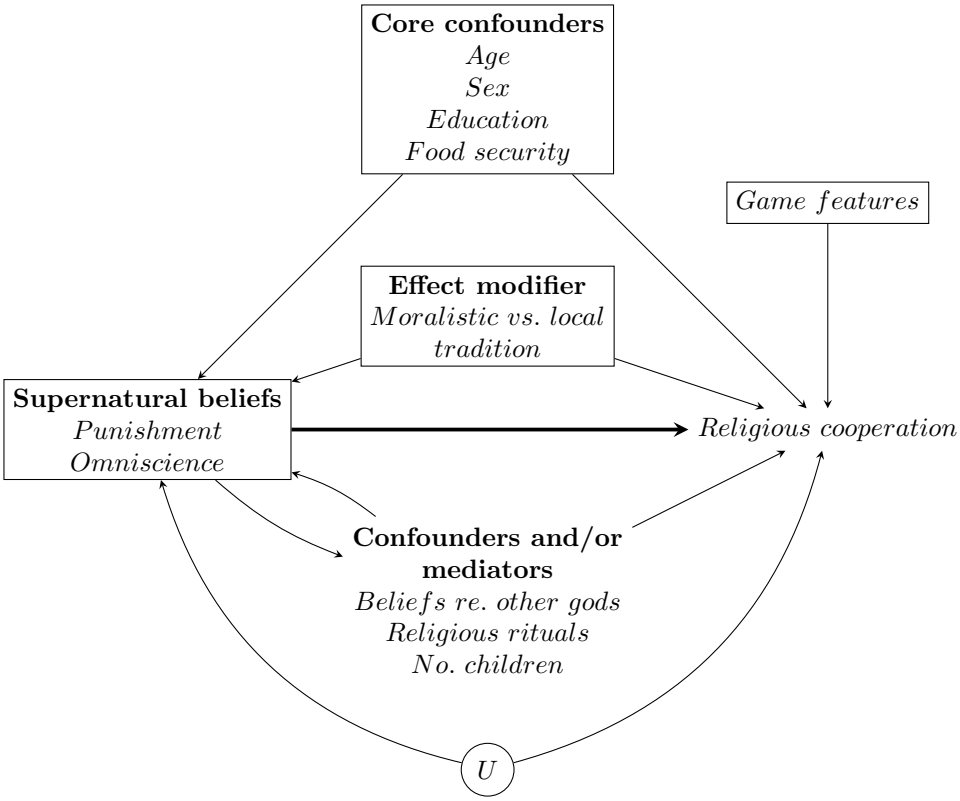

Figure 2: Directed Acyclic Graph (DAG) displaying our core causal assumptions regarding the relationship between supernatural beliefs (our exposure) and religious cooperation (our outcome).

each site as a random intercept. Exposures and covariates will be included as random slopes to allow these relationships to vary by site (i.e., a fully-varying random effects model). Note that we will not include the site-level effect modifier "moralistic vs. local" tradition as a covariate in these models; instead, we will analyse the model's posterior predictions by local vs. moralistic tradition to examine potential effect modification. We also include the mean punishment and omniscience scores for each site as group-level covariates, in order to account for potential unobserved group-level confounding between these exposures and our outcomes (i.e., a "Mundlak" model; Mundlak, 1978; Bendixen and Purzycki, 2026).

This base model will be repeated twice for each game, first in the SELF vs. DISTANT condition, and then in the LOCAL vs DISTANT condition. We will model RAG outcomes with an aggregated binomial distribution, and DGs with an ordinal distribution.

All analyses will be Bayesian models using Stan (Team, 2026) via the "brms" package (Bürkner, 2017). Each model will run with four chains for 2,000 iterations each, with 1,000 chains as warm-up. R-hat values close to 1.00, adequate effective sample sizes, and visual inspection of convergence plots between chains will be used to ensure that models are well-mixed and sufficiently powered (this will be updated as needed; e.g., increasing the number of sampling iterations). Weakly-regularising

priors will be used throughout. All analyses will be conducted in R version 4.4.1 (Team, 2024). For example code demonstrating our analysis approach on simulated data, see here.

G-computation methods (Hernán and Robins, 2020) will be used to estimate our marginal causal effects of interest; i.e., the joint effect of a change in supernatural punishment and omniscience from both mini-scales being "0" (no punishment/omniscience) to both being "1" (maximally punitive/omniscient), and whether this differs by local vs. moralistic religious tradition. These causal contrasts will be calculated based on the posterior predictions from the Bayesian models. As coefficients from binomial and ordinal models are not necessarily intuitive to interpret (i.e., odds ratios/log-odds), these marginal effects will reflect the average change in the outcome on the count scale (i.e., the difference in the number of donations to DISTANT). The structure of these core statistical models are given in Supplementary Section 1.6.1.

As noted above, in addition to this base model there are a range of different potential causal structures and other factors to consider. We will therefore conduct sensitivity analyses to explore how robust our results are to differing causal assumptions. These will be discussed briefly below, with more detail provided in Supplementary Section 1.6.2. We will assess:

1. Different operational definitions of moralistic vs. local religious traditions, as described above.
2. A multiplicative, rather than additive, effect between the punishment and omniscience exposures, to assess any synergistic effects between exposures.
3. Different assumptions regarding "confounders" vs. "mediators" for the variables beliefs about other gods, religious rituals towards the focal deity, and number of children (Figure 2).

Based on our previous experience with these data, there are some variables with small amounts of missing data across most of the sites (estimated <10% missing). Given the risk of potential selection bias and inefficient analyses, we will impute missing data to mitigate these concerns (for more details, see Supplementary Section 1.6.3). Imputed and complete-case results will be compared, to assess how/if results differ.

Ethics

For the most recently collected data, Aarhus University's Research Ethics Committee initially reviewed and approved the project. When it was an option, each field researcher subsequently got local approval at their own institution using the same protocol, and including Aarhus University's approval as an appendix.

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